UNIVERSIDAD PERUANA LOS ANDES

FACULTAD DE INGENIERIA

ESCUELA PROFECIONAL DE INGENERIA DE SISTEMAS Y COMPUTACIÓN

TEMA:

Informe Sobre Comandos De Protocolos De Red

CURSO: Transmisión de Datos

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CICLO: V

SECCION: A1

HUANCAYO - PERÚ

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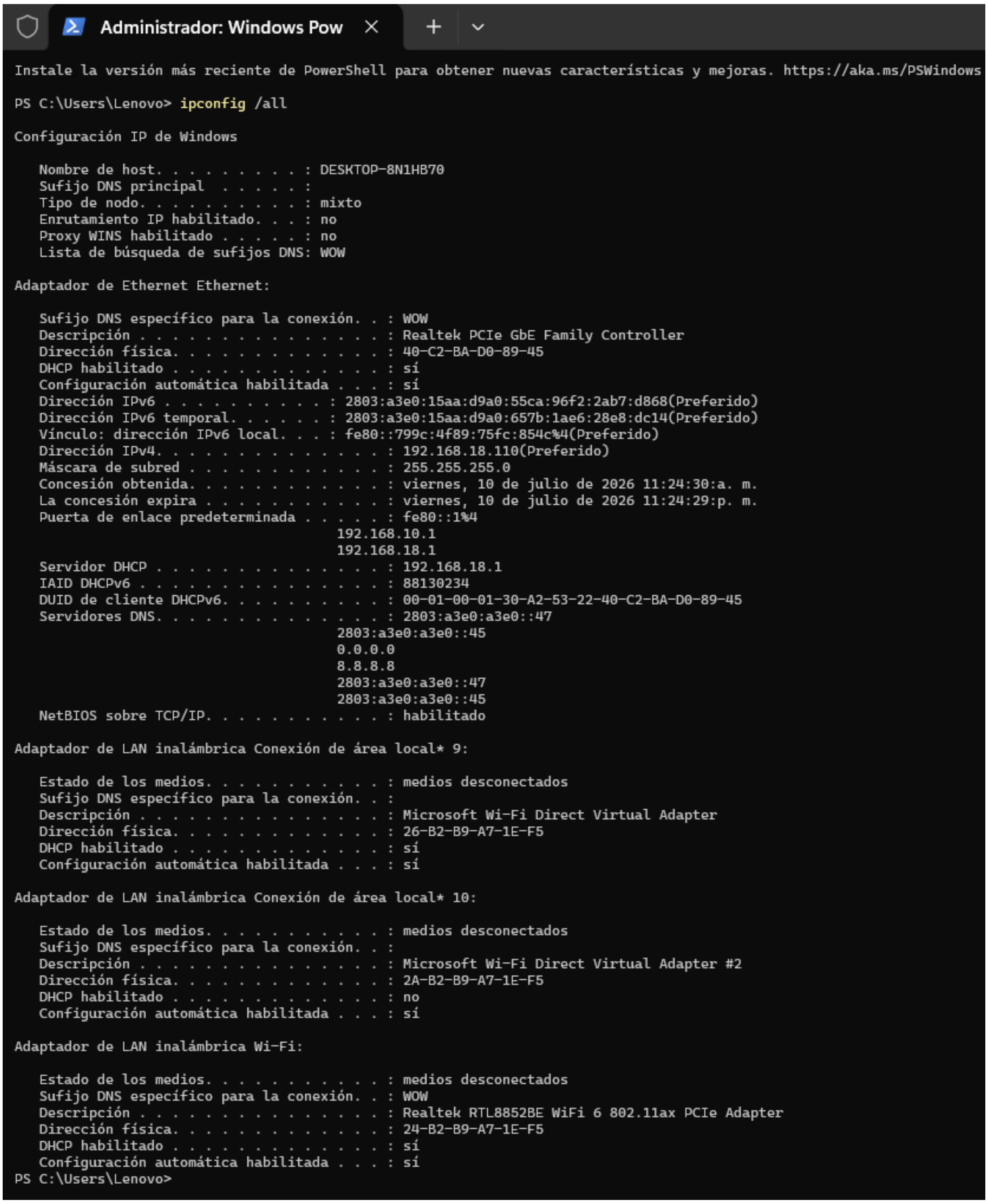
Índice:

Práctica de Comandos útiles en Windows/Powershell.

1. Ver configuración TCP/IP del PC:




```
Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

PS C:\Users\Lenovo> ipconfig /all

Configuración IP de Windows

   Nombre de host. . . . . . . . . : DESKTOP-8N1HB70
   Sufijo DNS principal  . . . . . :
   Tipo de nodo. . . . . . . . . . : mixto
   Enrutamiento IP habilitado. . . : no
   Proxy WINS habilitado . . . . . : no
   Lista de búsqueda de sufijos DNS: WOW

Adaptador de Ethernet Ethernet:

   Sufijo DNS específico para la conexión. . : WOW
   Descripción . . . . . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Dirección física. . . . . . . . . . . . . : 40-C2-BA-D0-89-45
   DHCP habilitado . . . . . . . . . . . . . : sí
   Configuración automática habilitada . . . : sí
   Dirección IPv6 . . . . . . . . . . : 2803:a3e0:15aa:d9a0:55ca:96f2:2ab7:d868(Preferido)
   Dirección IPv6 temporal. . . . . . : 2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14(Preferido)
   Vínculo: dirección IPv6 local. . . : fe80::799c:4f89:75fc:854c%4(Preferido)
   Dirección IPv4. . . . . . . . . . . . . . : 192.168.18.110(Preferido)
   Máscara de subred . . . . . . . . . . . . : 255.255.255.0
   Concesión obtenida. . . . . . . . . . . . : viernes, 10 de julio de 2026 11:24:30:a. m.
   La concesión expira . . . . . . . . . . . : viernes, 10 de julio de 2026 11:24:29:p. m.
   Puerta de enlace predeterminada . . . . . : fe80::1%4
                                       192.168.10.1
                                       192.168.18.1
   Servidor DHCP . . . . . . . . . . . . . . : 192.168.18.1
   IAID DHCPv6 . . . . . . . . . . . . . . . : 88130234
   DUID de cliente DHCPv6. . . . . . . . . . : 00-01-00-01-30-A2-53-22-40-C2-BA-D0-89-45
   Servidores DNS. . . . . . . . . . . . . . : 2803:a3e0:a3e0::47
                                       2803:a3e0:a3e0::45
                                       0.0.0.0
                                       8.8.8.8
                                       2803:a3e0:a3e0::47
                                       2803:a3e0:a3e0::45
   NetBIOS sobre TCP/IP. . . . . . . . . . . : habilitado

Adaptador de LAN inalámbrica Conexión de área local* 9:

   Estado de los medios. . . . . . . . . . . : medios desconectados
   Sufijo DNS específico para la conexión. . :
   Descripción . . . . . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Dirección física. . . . . . . . . . . . . : 26-B2-B9-A7-1E-F5
   DHCP habilitado . . . . . . . . . . . . . : sí
   Configuración automática habilitada . . . : sí

Adaptador de LAN inalámbrica Conexión de área local* 10:

   Estado de los medios. . . . . . . . . . . : medios desconectados
   Sufijo DNS específico para la conexión. . :
   Descripción . . . . . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Dirección física. . . . . . . . . . . . . : 2A-B2-B9-A7-1E-F5
   DHCP habilitado . . . . . . . . . . . . . : no
   Configuración automática habilitada . . . : sí

Adaptador de LAN inalámbrica Wi-Fi:

   Estado de los medios. . . . . . . . . . . : medios desconectados
   Sufijo DNS específico para la conexión. . : WOW
   Descripción . . . . . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Dirección física. . . . . . . . . . . . . : 24-B2-B9-A7-1E-F5
   DHCP habilitado . . . . . . . . . . . . . : sí
   Configuración automática habilitada . . . : sí
PS C:\Users\Lenovo>
```

2. Probar conectividad a un host:

```
PS C:\Users\Lenovo> ping google.com

Haciendo ping a google.com [2800:3f0:4005:419::200e] con 32 bytes de datos:
Respuesta desde 2800:3f0:4005:419::200e: tiempo=42ms
Respuesta desde 2800:3f0:4005:419::200e: tiempo=42ms
Respuesta desde 2800:3f0:4005:419::200e: tiempo=43ms
Respuesta desde 2800:3f0:4005:419::200e: tiempo=42ms

Estadísticas de ping para 2800:3f0:4005:419::200e:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 42ms, Máximo = 43ms, Media = 42ms
```

3. Ver la tabla de rutas:

```
PS C:\Users\Lenovo> route print
===========================================================================
ILista de interfaces
  4...40 c2 ba d0 89 45 ......Realtek PCIe GbE Family Controller
 18...26 b2 b9 a7 1e f5 ......Microsoft Wi-Fi Direct Virtual Adapter
 13...2a b2 b9 a7 1e f5 ......Microsoft Wi-Fi Direct Virtual Adapter #2
  7...24 b2 b9 a7 1e f5 ......Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Tabla de enrutamiento
===========================================================================
Rutas activas:
Destino de red        Máscara de red   Puerta de enlace   Interfaz  Métrica
          0.0.0.0          0.0.0.0     192.168.10.1   192.168.18.110    281
          0.0.0.0          0.0.0.0     192.168.18.1   192.168.18.110     25
        127.0.0.0        255.0.0.0      En vínculo         127.0.0.1    331
        127.0.0.1  255.255.255.255      En vínculo         127.0.0.1    331
  127.255.255.255  255.255.255.255      En vínculo         127.0.0.1    331
     192.168.18.0    255.255.255.0      En vínculo    192.168.18.110    281
   192.168.18.110  255.255.255.255      En vínculo    192.168.18.110    281
   192.168.18.255  255.255.255.255      En vínculo    192.168.18.110    281
        224.0.0.0        240.0.0.0      En vínculo         127.0.0.1    331
        224.0.0.0        240.0.0.0      En vínculo    192.168.18.110    281
  255.255.255.255  255.255.255.255      En vínculo         127.0.0.1    331
  255.255.255.255  255.255.255.255      En vínculo    192.168.18.110    281
===========================================================================
Rutas persistentes:
  Dirección de red  Máscara de red  Dirección de puerta de enlace  Métrica
          0.0.0.0          0.0.0.0     192.168.10.1  Predeterminada
===========================================================================

IPv6 Tabla de enrutamiento
===========================================================================
Rutas activas:
 Cuando destino de red métrica      Puerta de enlace
  4   4121 ::/0                     fe80::1
  1    331 ::1/128                  En vínculo
  4   4121 2803:a3e0:15aa:d9a0::/64 En vínculo
  4    281 2803:a3e0:15aa:d9a0:55ca:96f2:2ab7:d868/128
                                    En vínculo
  4    281 2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14/128
                                    En vínculo
  4    281 fe80::/64                En vínculo
  4    281 fe80::799c:4f89:75fc:854c/128
                                    En vínculo
  1    331 ff00::/8                 En vínculo
  4    281 ff00::/8                 En vínculo
===========================================================================
Rutas persistentes:
  Ninguno
```

4. Ver conexiones TCP activas:

```
PS C:\Users\Lenovo> netstat -an | findstr "TCP"
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49675          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:50131          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:57634          0.0.0.0:0              LISTENING
  TCP    127.0.0.1:1434         0.0.0.0:0              LISTENING
  TCP    127.0.0.1:9010         0.0.0.0:0              LISTENING
  TCP    127.0.0.1:9180         0.0.0.0:0              LISTENING
  TCP    127.0.0.1:45654        0.0.0.0:0              LISTENING
  TCP    127.0.0.1:59618        0.0.0.0:0              LISTENING
  TCP    127.0.0.1:59852        0.0.0.0:0              LISTENING
  TCP    127.0.0.1:63613        127.0.0.1:28194        SYN_SENT
  TCP    192.168.18.110:139     0.0.0.0:0              LISTENING
  TCP    192.168.18.110:49672   204.79.197.222:443     ESTABLISHED
  TCP    192.168.18.110:49674   40.79.150.120:443      TIME_WAIT
  TCP    192.168.18.110:52828   216.239.36.223:443     ESTABLISHED
  TCP    192.168.18.110:54334   20.42.179.192:443      TIME_WAIT
  TCP    192.168.18.110:54336   20.42.179.192:443      TIME_WAIT
  TCP    192.168.18.110:55772   13.107.6.254:443       ESTABLISHED
  TCP    192.168.18.110:58100   40.126.45.17:443       TIME_WAIT
  TCP    192.168.18.110:58102   51.104.15.253:443      TIME_WAIT
  TCP    [::]:135               [::]:0                 LISTENING
  TCP    [::]:445               [::]:0                 LISTENING
  TCP    [::]:49664             [::]:0                 LISTENING
  TCP    [::]:49665             [::]:0                 LISTENING
  TCP    [::]:49666             [::]:0                 LISTENING
  TCP    [::]:49667             [::]:0                 LISTENING
  TCP    [::]:49668             [::]:0                 LISTENING
  TCP    [::]:49675             [::]:0                 LISTENING
  TCP    [::]:50131             [::]:0                 LISTENING
  TCP    [::1]:1434             [::]:0                 LISTENING
  TCP    [::1]:42050            [::]:0                 LISTENING
  TCP    [::1]:59618            [::]:0                 LISTENING
  TCP    [::1]:59852            [::]:0                 LISTENING
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:49152  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:49540  [2603:1030:40c:e::]:443  ESTABLISHED
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:49544  [2603:1030:40c:e::]:443  ESTABLISHED
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:49669  [2607:f8b0:400c:c12::bc]:5228  ESTABLISHED
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:49670  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:49671  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:49677  [2600:1901:1:7c5::]:443  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:49812  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:50157  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:50221  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:50381  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:50519  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:50537  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:50554  [2603:1063:2001:1812::365:ff1]:443  ESTABLISHED
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:50587  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:51078  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:51215  [2803:a3e0:a3e0::45]:53  TIME_WAIT
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:51343  [2603:1063:2001:1805::365:ff1]:443  ESTABLISHED
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:51355  [2803:a3e0:a3e0::45]:53  TIME_WAIT
```

```
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:51481  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:53084  [2603:1056:c04:c22::2]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:53111  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:53436  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:53834  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:54331  [2607:6bc0::10]:443    ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:54469  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:54586  [2620:1ec:d::254]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:54596  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:54614  [2a03:2880:f22c:1c5:face:b00c:0:167]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:54964  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:55058  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:55330  [2600:1419:3200:289::57]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:55652  [2603:1061:13f5:c0c::365:590]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:55881  [2a04:4e42:36::684]:80  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:56261  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:56596  [2600:1419:3200::be62:a11a]:443  CLOSE_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:56597  [2800:240:e:5::b819:7412]:443  CLOSE_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:56645  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:56700  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:56963  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:57318  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:57402  [2600:1901:0:179c::]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:57514  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:57743  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:57822  [2800:3f0:4005:404::200e]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:58101  [2603:1056:c04:c21::2]:443  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:58137  [2600:1901:0:179c::]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:58138  [2001:4860:4860::8844]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:58493  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:58544  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:58604  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:59023  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:59200  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:59317  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:59740  [2603:1036:2401:1::14]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:59748  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:60562  [2800:3f0:4005:419::2003]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:60724  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:60878  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:61008  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:61549  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:61695  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:61926  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:62276  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:62419  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:62547  [2603:1063:2001:1805::365:ff1]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:62665  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:63090  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:63139  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:63611  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:63893  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:64018  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:64125  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:64308  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:64380  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:64416  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:64785  [2603:1063:2001:180d::365:ff1]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:64807  [2603:1063:2000::12]:443  ESTABLISHED
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:64855  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:65077  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:65202  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:65210  [2803:a3e0:a3e0::45]:53  TIME_WAIT
TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:65526  [2603:1020:c04:8::691]:443  ESTABLISHED
```

5. Probar conexión TCP a un puerto específico:

```
PS C:\Users\Lenovo> Test-NetConnection -ComputerName google.com -Port 443


ComputerName     : google.com
RemoteAddress    : 2800:3f0:4005:40c::200e
RemotePort       : 443
InterfaceAlias   : Ethernet
SourceAddress    : 2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14
TcpTestSucceeded : True
```

Práctica de Comandos UDP.

1. Ver conexiones UDP activas:

```
PS C:\Users\Lenovo> netstat -an | findstr "UDP"
  UDP    0.0.0.0:500            *:*
  UDP    0.0.0.0:4500           *:*
  UDP    0.0.0.0:5050           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5355           *:*
  UDP    0.0.0.0:49744          0.0.32.14:443
  UDP    0.0.0.0:51069          *:*
  UDP    0.0.0.0:51875          0.0.0.16:443
  UDP    0.0.0.0:52874          3.101.15.241:443
  UDP    0.0.0.0:53058          *:*
  UDP    0.0.0.0:53778          0.0.32.3:443
  UDP    0.0.0.0:57633          0.0.32.10:443
  UDP    0.0.0.0:58010          *:*
  UDP    0.0.0.0:58527          0.0.32.14:443
  UDP    0.0.0.0:60488          0.0.32.10:443
  UDP    0.0.0.0:62975          *:*
  UDP    0.0.0.0:63832          0.0.0.16:443
  UDP    127.0.0.1:1900         *:*
  UDP    127.0.0.1:10280        *:*
  UDP    127.0.0.1:10281        *:*
  UDP    127.0.0.1:49664        127.0.0.1:49664
  UDP    127.0.0.1:51053        *:*
  UDP    127.0.0.1:65315        *:*
  UDP    192.168.18.110:137     *:*
  UDP    192.168.18.110:138     *:*
  UDP    192.168.18.110:1900    *:*
  UDP    192.168.18.110:2177    *:*
  UDP    192.168.18.110:51052   *:*
  UDP    [::]:500               *:*
  UDP    [::]:4500              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5355              *:*
  UDP    [::]:49744             [2800:3f0:4005:400::200e]:443
  UDP    [::]:51069             [2001:4860:4826:7700::]:443
  UDP    [::]:51875             [2607:6bc0::10]:443
  UDP    [::]:52874             [2603:1063:2001:180a::365:ff1]:443
  UDP    [::]:53058             *:*
  UDP    [::]:53778             [2800:3f0:4005:40a::2003]:443
  UDP    [::]:57633             [2800:3f0:4005:404::200a]:443
  UDP    [::]:58010             *:*
  UDP    [::]:58527             [2800:3f0:4005:404::200e]:443
  UDP    [::]:60488             [2800:3f0:4005:40c::200a]:443
  UDP    [::]:62975             *:*
  UDP    [::]:63832             [2607:6bc0::10]:443
  UDP    [::1]:1900             *:*
  UDP    [::1]:51051            *:*
  UDP    [2803:a3e0:15aa:d9a0:55ca:96f2:2ab7:d868]:2177  *:*
  UDP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:2177  *:*
  UDP    [fe80::799c:4f89:75fc:854c%4]:1900  *:*
  UDP    [fe80::799c:4f89:75fc:854c%4]:2177  *:*
  UDP    [fe80::799c:4f89:75fc:854c%4]:51050  *:*
```

2. Probar puerto UDP (ejemplo: DNS 53):

```
PS C:\Users\Lenovo> $PSVersionTable.PSVersion

Major  Minor  Build  Revision
-----  -----  -----  --------
5      1      26100  8737


PS C:\Users\Lenovo> Test-NetConnection -ComputerName 8.8.8.8 -Port 53 -InformationLevel Detailed


ComputerName             : 8.8.8.8
RemoteAddress            : 8.8.8.8
RemotePort               : 53
NameResolutionResults    : 8.8.8.8
                           dns.google
MatchingIPsecRules       :
NetworkIsolationContext  : Internet
InterfaceAlias           : Ethernet
SourceAddress            : 192.168.18.110
NetRoute (NextHop)       : 192.168.18.1
TcpTestSucceeded         : True
```

3. Enviar paquetes UDP:

```
PS C:\Windows\System32> $udpClient = New-Object System.Net.Sockets.UdpClient
>> $bytes = [System.Text.Encoding]::ASCII.GetBytes("test")
>> $udpClient.Send($bytes, $bytes.Length, "192.168.1.10", 5000)
>> $udpClient.Close()
4
```

Práctica HTTP en Windows/Powershell:

1. Realizar petición GET:

```
PS C:\Windows\System32> Invoke-WebRequest -Uri "https://example.com"

StatusCode        : 200
StatusDescription : OK
Content           : <!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon"
                    href="data:,"><meta name="viewport" content="width=device-width,
                    initial-scale=1"><style>body{background:#eee;width:6…
RawContent        : HTTP/1.1 200 OK
                    Date: Fri, 10 Jul 2026 17:49:20 GMT
                    Transfer-Encoding: chunked
                    Connection: keep-alive
                    Server: cloudflare
                    Age: 7659
                    Cf-Cache-Status: HIT
                    CF-RAY: a191678ccf2a1eab-EZE
                    Content-Typ…
Headers           : {[Date, System.String[]], [Transfer-Encoding, System.String[]], [Connection, System.String[]],
                    [Server, System.String[]]…}
Images            : {}
InputFields       : {}
Links             : {@{outerHTML=<a href="https://iana.org/domains/example">Learn more</a>; tagName=A;
                    href=https://iana.org/domains/example}}
RawContentLength  : 559
RelationLink      : {}
```

2. Descargar un archivo:

```
PS C:\Windows\System32> Invoke-WebRequest -Uri "https://www.w3.org/WAI/ER/tests/xhtml/testfiles/resources/pdf/dummy.pdf" -OutFile "file.pdf"
PS C:\Windows\System32>
```

3. Ver encabezados HTTP:

```
PS C:\Windows\System32> Invoke-WebRequest -Uri "https://google.com" -Method Head

StatusCode        : 200
StatusDescription : OK
Content           :
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-hgx64aKK32_1vbZdF2X2Yg'
                    'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: …
Headers           : {[Content-Security-Policy-Report-Only, System.String[]], [Accept-CH, System.String[]], [P3P, System.String[]], [Date,
                    System.String[]]…}
Images            : {}
InputFields       : {}
Links             : {}
RawContentLength  : 0
RelationLink      : {}
```

4. Revisar puertos HTTP activos:

```
PS C:\Windows\System32> netstat -an | findstr ":80"
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:55563  [2603:1056:c04:802::2]:443  ESTABLISHED
  TCP    [2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14]:55564  [2603:1056:c04:802::2]:443  ESTABLISHED
PS C:\Windows\System32>
```

5. Probar conexión HTTP:

```
PS C:\Windows\System32> Test-NetConnection -ComputerName example.com -Port 80

ComputerName     : example.com
RemoteAddress    : 2606:4700:10::6814:179a
RemotePort       : 80
InterfaceAlias   : Ethernet
SourceAddress    : 2803:a3e0:15aa:d9a0:657b:1ae6:28e8:dc14
TcpTestSucceeded : True

PS C:\Windows\System32>
```

Práctica FTP en Windows:

1. Iniciar cliente FTP:

```
PS C:\Windows\System32> ftp
ftp>
```

2. Conectarse a un servidor FTP:

```
PS C:\Users\Usuario> ftp ftp.debian.org
Conectado a ftp.debian.org.
220 ftp.debian.org FTP server (vsftpd)
Usuario (ftp.debian.org:(none)): anonymous
331 Please specify the password.
Contraseña:
230 Login successful.
ftp>
```

3. Comandos FTP más usados:

ls:

```
ftp> ls
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
-rw-r--r--    1 1176     1176         2510 Jan 14  2022 README
-rw-r--r--    1 1176     1176         1848 Jan 14  2022 README.html
drwxr-xr-x    3 1176     1176         4096 Jan 14  2022 debian
drwxr-xr-x    2 1176     1176         4096 Jan 14  2022 pub
drwxr-xr-x    2 1176     1176         4096 Jan 14  2022 tools
226 Directory send OK.
ftp>
```

cd:

```
ftp> cd pub
250 Directory successfully changed.
ftp>
```

get file:

```
ftp> get README
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for README (2510 bytes).
226 Transfer complete.
2510 bytes recibidos en 0,00 segundos (6,12 MB/s)
ftp>
```

bye:

```
ftp> bye
221 Goodbye.
PS C:\Users\Usuario>
```

4. Probar puerto FTP:

```
PS C:\Users\Usuario> Test-NetConnection -ComputerName ftp.debian.org -Port 21


ComputerName           : ftp.debian.org
RemoteAddress          : 130.89.148.12
RemotePort             : 21
InterfaceAlias         : Ethernet
SourceAddress          : 192.168.1.100
PingSucceeded          : True
PingReplyDetails (RTT) : 24 ms
TcpTestSucceeded       : True
```

5. Ver tráfico FTP en la máquina:

```
PS C:\Users\Usuario> netstat -an | findstr ":21"
  TCP    192.168.1.100:54321    130.89.148.12:21       ESTABLISHED
  TCP    192.168.1.100:54322    130.89.148.12:21       TIME_WAIT
  TCP    0.0.0.0:21             0.0.0.0:0              LISTENING
  TCP    [::]:21                [::]:0                 LISTENING
```

Práctica de SMTP en PowerShell:

1. Probar conexión SMTP:

```
PS C:\Users\Usuario> Test-NetConnection -ComputerName smtp.gmail.com -Port 587


ComputerName           : smtp.gmail.com
RemoteAddress          : 142.250.185.109
RemotePort             : 587
InterfaceAlias         : Ethernet
SourceAddress          : 192.168.1.100
PingSucceeded          : True
PingReplyDetails (RTT) : 18 ms
TcpTestSucceeded       : True
```

2. Enviar correo (requiere credenciales):

```
PS C:\Users\Usuario> Send-MailMessage -From "tucorreo@example.com" -To "destino@example.com" -Subjec
t "Prueba" -Body "Hola, esto es un test" -SmtpServer smtp.gmail.com -Port 587
```

```
Send-MailMessage : El servidor SMTP requirió una autenticación o las credenciales no son válidas.
La respuesta del servidor fue: 530-5.5.1 Authentication Required. Learn more at
530 5.5.1 https://support.google.com/mail/?p=WantAuthError 122-20020a170902ee8b00b001f8b4e2d6ddsm123
4567plb.23 - gsmtp
En línea: 1 Carácter: 1
+ Send-MailMessage -From "tucorreo@example.com" -To "destino@example.com ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidOperation: (System.Net.Mail.SmtpClient:SmtpClient) [Send-MailMe
ssage], SmtpException
    + FullyQualifiedErrorId : SmtpException,Microsoft.PowerShell.Commands.SendMailMessage
```

3. Ver si hay conexiones SMTP en la PC:

```
PS C:\Users\Usuario> netstat -an | findstr ":25"
  TCP    192.168.1.100:54321    142.250.185.109:25    TIME_WAIT
  TCP    192.168.1.100:54322    142.250.185.109:25    ESTABLISHED
  TCP    0.0.0.0:25             0.0.0.0:0             LISTENING
  TCP    [::]:25                [::]:0                LISTENING
```